T3 Firmware & Bootloader

Trying the commands · built Nov 13, 2026

Trying the commands

The sequence to send by hand, in order, when you want to watch an update happen rather than trust it.

12. Testing Modbus Commands

Modbus RTU (Serial) Example Commands

TEXT

```
Python pseudo-code using Modbus:

# 1. Set expected MD5 checksum (registers 1995-1998)
write_registers(1995, [0xABCD, 0xEF12, 0x3456, 0x7890])

# 2. Initiate firmware update (START)
write_register(16, 0x1F)  # UPDATE_STATUS = 0x1F

# 3. Send firmware packets via protocol
# (In serial/TCP stream, not Modbus)

# 4. Check resume position if interrupted
result = read_registers(17, 2)  # READ UPDATE_PTR_HI/LO

# 5. Resume from last position (if needed)
write_register(16, 0x1F)  # UPDATE_STATUS = 0x1F (resume)

# 6. Complete update
write_register(16, 0x01)  # UPDATE_STATUS = 0x01

# 7. Read completion status
status = read_register(23)  # UPDATE_FLAG
if status == 0x55:
    print("Update successful!")
else:
    print("Update failed!")

# 8. Full erase if needed
write_register(16, 0x3F)  # UPDATE_STATUS = 0x3F (erase all)
```